LI Weiyi

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EDUCATION

Sichuan University **Chengdu, China | 09/2023 – 06/2027**
Light Chemical Engineering Weighted Average Mark: 86.19%
Core modules: Experiment for Engineering Chemistry, Physics Experiments, Chemical Engineering Graphics, Physical Chemistry, Principle of Chemical Engineering, Polymer Chemistry, Fundamentals of Biomaterials and Biomedical Devices, etc.

RESEARCH EXPERIENCE

Summer Research Programme, Westlake University **04/2026 – 08/2026**
- Mastered the one-pot preparation principle of PBFDO through the combination of oxidative polymerization and reduction doping, became familiar with the novel catalyst-free rapid synthesis method in a DMSO/acetic anhydride mixed system, understood the directions for improving the synthesis process, and explored the combined applications of PBFDO with biomass materials;
- Gained an in-depth understanding of the preparation mechanism of MXene materials, mastered their basic synthesis methods, and learned about the current research progress and mainstream application scenarios of the material.

High-Strength, Transparent, Antibacterial, Anti-Freezing, Water-Retaining, Polyacrylamide/Poly(itaconic acid)-Based Conductive Hydrogel Reinforced by Tanned Collagen Fiber Network for Flexible Sensing and Triboelectric Nanogenerator Applications, *Colloids and Surfaces A: Physicochemical and Engineering Aspects (Major revision, Second author)*
06/2025 – 01/2026
- Developed a polyacrylamide/polyitaconic acid-based conductive hydrogel (PAIAH) using tanned goatskin collagen fiber network as a robust skeleton and aluminum chloride hexahydrate as a conductive medium, achieving synergistic optimization of high strength (tensile strength of 24.4 MPa) and transparency (73.4% transmittance at 550 nm) with freeze resistance down to -20°C;
- Constructed an interpenetrating network structure between collagen fiber and polymer networks via monomer impregnation followed by in-situ thermal polymerization, enabling the material to combine 71.5% elongation at break, strain-responsive properties with a sensitivity coefficient of 3.24, and stable electrical conductivity under 600-second cyclic loading;
- Demonstrated 84.3% weight retention after 15 days in a 30°C/40% relative humidity environment, along with antibacterial activity against Escherichia coli and Staphylococcus aureus, expanding applications for biocompatible electronic materials;
- Monitored human motions (facial expressions, joint flexion) and bioelectrical signals (electrocardiogram, electromyogram), and enabled self-powered applications such as nighttime safety alerts for visually impaired individuals when integrated into triboelectric nanogenerators;
- Integrated the mechanical durability of natural collagen fibers with the functional versatility of hydrogels to provide a sustainable material solution for wearable health monitoring and energy harvesting applications.

Self-adhesive, Anti-freezing, Multifunctional PDES-Based Conductive Hydrogel Strengthened by Collagen Fiber Network for Multimodal Sensing and Triboelectric Nanogenerator, *Chemical Engineering Journal (Accepted, First author)*
01/2025 – 01/2026
- Developed a composite conductive hydrogel (DPAMG) using a goatskin collagen fiber network (GCFN) as the skeleton and polymerizable deep eutectic solvent (PDES) as the conductive medium, to address the performance limitations of flexible wearable devices in complex scenarios, achieving synergistic optimization of high mechanical strength (maximum stress of 3.3 MPa, elongation at break of 500%) and high ionic conductivity (5.14 S/m);
- Constructed an interpenetrating collagen fiber-polymer network via in-situ polymerization, and integrated a crosslinking system of sodium phytate and N,N'-methylenebisacrylamide, endowing DPAMG with freeze resistance, moisture retention, antibacterial properties, and network stability while overcoming the environmental adaptability limitations of traditional hydrogels;
- Enabled DPAMG to support multimodal sensing for precise monitoring of physiological signals such as joint movements, facial expressions, electrocardiograms, and electromyograms, and functioned as a core component of single-electrode triboelectric nanogenerators (S-TENGs) to output stable electrical energy for driving LEDs, thereby expanding applications in self-powered flexible electronics;
- Provided a novel paradigm for high-value utilization of animal skin collagen fibers, resolved the strength-conductivity trade-off in deep eutectic solvent-based conductive hydrogels, and offered an innovative material platform for integrating flexible electronics with sustainable energy systems.

Gelatin-Based Conductive Hydrogels Accelerate Diabetic Chronic Wound Healing by Regulating NF-kB and PI3K/Akt/mTOR Pathways **01/2025 – 01/2026**
- Developed a dual-drug-loaded gelatin-based conductive hydrogel (HGP@D) based on dual dynamic covalent cross-linking of imine bonds and boronic ester bonds, which possesses various excellent properties including superior mechanical properties and good tissue adhesion strength;
- Enabled the HGP@D hydrogel to exhibit glucose/pH dual-responsive drug release behavior, possess biomimetic electroactivity, potent broad-spectrum antibacterial activity, and significant intracellular antioxidant activity, and promoted the migration of fibroblasts and endothelial cells;
- Made the hydrogel regulate the NF-κB pathway to induce macrophage polarization toward the M2 type, facilitating the transition of wounds from the inflammatory phase to the proliferative phase, and activated the PI3K/Akt/mTOR pathway through a continuous endogenous electric field coupling effect to promote angiogenesis;
- Achieved synergistic immunomodulation through the combined anti-inflammatory and pro-angiogenic effects of the HGP@D hydrogel, effectively promoting the healing of full-thickness diabetic chronic wounds infected with Staphylococcus aureus.

Research on Uranium Extraction from Seawater **12/2024 – 09/2025**
- Utilized plasma-induced grafting technology to prepare amidoxime-functionalized carbon black adsorbent (CB-AO) for uranium extraction from seawater, and determined the optimal plasma treatment conditions during the material preparation stage, working out CB-AO with the retention of the basic structure of carbon black, an increased surface functional group density, and improved hydrophilicity;
- Employed multiple techniques during the characterization and performance testing stage to confirm group grafting and structure, receiving the results that this material exhibited a static adsorption capacity of 220.95 mg/g, a dynamic simulated seawater adsorption capacity of 3.2 mg/g, and demonstrated good selectivity;

- Revealed the bidentate chelation mechanism between nitrogen and oxygen atoms and uranium ions during the mechanism and data integration stage, which was a spontaneous endothermic reaction with a capacity retention rate of 81.4% after 4 cycles, and simultaneously, issues such as a decrease in adsorption capacity during long-term high-salt operation were identified;
- Addressed the research issues by planning to advance in three aspects in the future: optimizing material performance, improving the preparation process, and deepening mechanism research, aiming to facilitate the industrialization of uranium extraction technology from seawater to alleviate the shortage of uranium resources.

Tissue Regeneration with Human-Derived Extracellular Matrix: Supporting Postoperative Repair in Oral Cancer, *Silver Award in the Internet+ Competition* **10/2024 – 09/2025**
- Participated in the background research module, systematically searched and sorted out relevant studies relying on authoritative academic databases such as Web of Science, laying a solid theoretical foundation for the project;
- Took charge of constructing the technical route, precisely identified the key and necessary application area of using human-derived extracellular matrix microneedle products for post-surgical repair of oral cancer based on the results of preliminary literature research;
- Engaged in the visual presentation of project outcomes, participated in designing promotional brochures to display project results in an intuitive and effective manner, facilitating the dissemination and promotion of the project.

Construction of Lipoic Acid Microgels and Their Application in Osteoarthritis Therapy **09/2024 – 09/2025**
- Spearheaded the development of lipoic acid microgels (LAGs) for osteoarthritis treatment, enabling targeted drug delivery to the joint cavity via injection and providing an innovative carrier solution for disease management;
- Designed a micron-scale LAGs structure that combines long-term retention in the joint cavity with rolling lubrication at the cartilage interface, effectively reducing cartilage wear while achieving sustained release of lipoic acid (LA);
- Leveraged the biodegradable properties of the LA-polymerized network to regulate degradation rates, enabling continuous release of anti-inflammatory and antioxidant components that significantly reduce joint inflammation and reactive oxygen species levels, thereby delaying disease progression.

Research on an Injectable Collagen-Based Hydrogel with Intelligent Wound pH Sensing, Antibacterial, and Antioxidant Properties **09/2024 – 09/2025**
- Applied selenium-rich tea leaves as raw materials and employ the pyrolysis method to prepare carbon dots, and oxidized chitosan with sodium periodate into dialdehyde chitosan to serve as a cross-linking agent;
- Took advantage of the Schiff base cross-linking reaction between the dialdehyde groups of chitosan and the amino groups of collagen, as well as the electrostatic interaction between carbon dots and dialdehyde chitosan, to form uniformly distributed carbon dot-dialdehyde chitosan imine bonds, and constructed an injectable composite hydrogel with a three-dimensional network structure;
- Enabled the injectable composite hydrogel to possess pH fluorescence response properties and effective anti-inflammatory functions due to the internal carbon dots, with chitosan endowing it with broad-spectrum antibacterial and hemostatic functions, while collagen enhancing its biocompatibility;
- Provided theoretical and data support for the clinical application of integrated diagnosis and treatment, such as monitoring the healing quality of diabetic chronic wounds in real-time during treatment, and applying in fields such as biosensing, wound healing, and antibacterial protection.

Transparent materials derived from sheepskin for light conversion application, *Applied Materials Today (Accepted, Second author)* **11/2023 – 06/2025**
- Constructed skin-based transparent materials using the three-dimensional network structure of zirconium-tanned sheepskin collagen fibers as the substrate, and infused specific monomers into the substrate via the dip-coating method and formed collagen fiber skeleton-based transparent materials (CFSTM) through in-situ free radical polymerization, with the polymer network densely filling the collagen fiber network;
- Ensured that CFSTM possesses high light transmittance (75.13% at 550 nm), excellent mechanical strength, good flexibility, antibacterial properties, and environmental stability;
- Added collagen-derived carbon quantum dots to the dipping solution to prepare CFSFTM with light conversion properties, made CFSFTM emit intense blue fluorescence under 365 nm ultraviolet light and exhibit excellent ultraviolet-visible light conversion performance, and prepared hydrophobic CFSFTM-W, ensuring good environmental stability in terms of transparency and fluorescence performance.

PRACTICAL EXPERIENCE

Hangzhou Gandashi Education Technology Co., Ltd. **08/2024 – 02/2025**
- Conducted learning method explanations for high school students, helping them master efficient learning strategies and improve learning efficiency;
- Formulated and supervised the implementation of daily study plans for high school students, adjusted the plan content according to individual student circumstances to ensure the orderly completion of learning tasks, and cultivated students' good study habits and time management skills;
- Communicated with students and their parents about study progress, provided feedback on learning situations, offered targeted suggestions, assisted students in improving their academic performance, and gained high recognition from both students and parents.

ADDITIONAL INFORMATION

Honors & Awards: Third-Class School Scholarship (2024-2025), Outstanding Student (2024-2025), Outstanding Team Leader in the School Practical Activity (05/2024)
Personal Interests: Cross-Disciplinary Humanities Podcast Listening, Music
Language: IELTS 6.5 (Listening 6.5, Reading 7.5, Writing 5.5, Speaking 6.5)